

● HARD

● CRAFT AND STRUCTURE

Craft and Structure

10 questions · ~15 min

09

Q1

CRAFT AND STRUCTURE

Mathematician Claude Shannon is widely regarded as a foundational figure in information theory. His most important paper, “A Mathematical Theory of Communication,” published in 1948 when he was employed at Bell Labs, utilized a concept called a “binary digit” (shortened to “bit”) to measure the amount of information in any signal and determine the fastest rate at which information could be transmitted while still being reliably decipherable. Robert Gallagher, one of Shannon’s colleagues, said that the bit was “[Shannon’s] discovery, and from it the whole communications revolution has sprung.”

Which choice best describes the overall structure of the text?

- A. It presents a theoretical concept, illustrates how the name of the concept has changed, and shows how the name has entered common usage.
- B. It introduces a respected researcher, describes an aspect of his work, and suggests why the work is historically significant.
- C. It names the company where an important mathematician worked, details the mathematician’s career at the company, and provides an example of the recognition he received there.
- D. It mentions a paper, offers a summary of the paper’s findings, and presents a researcher’s commentary on the paper.

While recent scholarship has undermined claims that the works of twelfth-century Islamic philosopher Ibn Rushd were _____ blank other Muslim philosophers of his time, it is indisputable that his location in the Muslim-ruled area of what is now Spain meant that his works were primarily available thousands of miles west of the era's center of Islamic thought.

Which choice completes the text with the most logical and precise word or phrase?

- A. controversial among
- B. antagonistic toward
- C. imitated by
- D. inconsequential to

The following text is from George Marion McClellan's 1895 poem "Eternity."

My spirit swoons, and all my senses cry
For Ocean's breast and covering of the sky.
Rock me to sleep, ye waves, and outward bound,
Just let me drift far out from toil and care,
Where lapping of the waves shall be the sound,
Which mingled with the winds that gently bear
Me on between a peaceful sea and sky,
To make my soothing slumberous lullaby.

Which choice best states the main purpose of the text?

- A. To illustrate the increasing intensity of the speaker's desire to escape ongoing hardship by gliding on the ocean
- B. To contrast the demands of the speaker's everyday life with the serenity of being rocked to sleep by the ocean
- C. To convey the speaker's longing for the ocean to impart a sense of inner tranquility
- D. To justify the speaker's qualms about being transported by the ocean to a quiet destination

Archaeologists have identified a plethora of beads fashioned from *Tritia gibbosula* shells in many Middle Stone Age (MSA) sites across western North Africa, including El Mnasra Cave, Morocco. In a 2021 paper, El Mehdi Sehasseh et al. attribute these artifacts' apparent _____ blank to the evolution and propagation of symbolic behavior (e.g., use of personal ornaments) in humans and the widespread availability of *T. gibbosula* during the MSA.

Which choice completes the text with the most logical and precise word or phrase?

- A. heterogeneity
- B. ubiquity
- C. expediency
- D. fecundity

Individual elephants and Arctic herbivores such as caribou tend to have fixed geographic ranges throughout their lifetimes, which had prompted some researchers to speculate that the Arctic woolly mammoth, an extinct elephantid, might have exhibited similar behavior. Mammoth tusks grew in sequential layers, incorporating ingested minerals and organics, and so each ivory stratum reflects the ratio of strontium isotopes ($^{87}\text{Sr}/^{86}\text{Sr}$) in the local environment; thus, the sequence of strata shows where the animal roamed during life. Recent analysis of the strontium ratios in the strata of one Arctic woolly mammoth tusk in relation to the geographic distribution of strontium ratios in the environment shows the animal's range begin to expand as it reached sexual maturity, only to contract again in its final 1.5 years.

Which choice best describes the function of the underlined statement in the text as a whole?

- A. It discusses a characteristic shared by certain animals in order to explain why researchers raised a possibility that turned out not to be supported by data described later in the text.
- B. It illustrates a pattern of behavior among certain animals in order to present a theory about exceptions to that pattern that is weakened by a finding described later in the text.
- C. It describes a similarity in the behavior of certain animals in order to show why a method described later in the text did not reveal whether another animal also showed that behavior.
- D. It introduces a trait shared by certain animals in order to contextualize a hypothesis about the origin of that trait that is advanced later in the text.

Seminole/Muscogee director Sterlin Harjo _____ blank television's tendency to situate Native characters in the distant past: this rejection is evident in his series *Reservation Dogs*, which revolves around teenagers who dress in contemporary styles and whose dialogue is laced with current slang.

Which choice completes the text with the most logical and precise word or phrase?

- A. repudiates
- B. proclaims
- C. foretells
- D. recants

A number of Indigenous politicians have been elected to the United States Congress since 2000 as members of the country's two established political parties. In Canada and several Latin American countries, on the other hand, Indigenous people have formed their own political parties to advance candidates who will advocate for the interests of their communities. This movement has been particularly successful in Ecuador, where Guadalupe Llori, a member of the Indigenous party known as Pachakutik, was elected president of the National Assembly in 2021.

Which choice best states the main purpose of the text?

- A. To trace the history of an Indigenous political movement and speculate about its future development
- B. To argue that Indigenous politicians in the United States should form their own political party
- C. To highlight two approaches to achieving political representation for Indigenous people
- D. To consider how Indigenous politicians in the United States have influenced Indigenous politicians in Canada and Latin America

As discussed by scholar Anna Mladentseva, many artworks produced in the mid-1990s to the early 2000s exclusively for exhibition on the internet, such as Sinae Kim's *Genesis* (2001), have become inaccessible because viewing them requires the use of _____ blank software (most notably Adobe Flash, discontinued in 2021).

Which choice completes the text with the most logical and precise word or phrase?

- A. defunct
- B. arcane
- C. ubiquitous
- D. extraneous

Vadamalai Elangovan and Ganapathi Marimuthu showed that high moonlight intensity inhibits the activity of the greater short-nosed fruit bat (*Cynopterus sphinx*), a result explicable in terms of benefits and costs: greater lunar intensity may not enable the bats to increase foraging success enough to offset the higher chance of detection by predatory owls or hawks. Most other nocturnal mammals respond to lunar intensity variations similarly to greater short-nosed fruit bats, but mongoose lemurs (*Eulemur mongoz*) display the opposite pattern, as their heavy reliance on visual foraging results in a different balance of reward and risk.

Which choice best describes the overall structure of the text?

- A. It discusses two different responses to the same natural phenomenon, explains why one of those responses occurs, and then suggests that the other response still needs to be explained.
- B. It describes and accounts for a finding, characterizes the finding as representative of a general pattern, and then describes and accounts for an exception to that pattern.
- C. It presents and explains a study result, indicates that the result is similar to the results of many other studies, and then attributes a conflicting study result to a difference in that study's methods.
- D. It introduces an observation of a behavioral pattern, presents an explanation for the pattern, and then describes an exception to the pattern that casts doubt on that explanation.

Whether the reign of a French monarch such as Hugh Capet or Henry I was historically consequential or relatively uneventful, its trajectory was shaped by questions of legitimacy and therefore cannot be understood without a corollary understanding of the factors that allowed the monarch to _____ blank his right to hold the throne.

Which choice completes the text with the most logical and precise word or phrase?

- A. disengage
- B. annotate
- C. buttress
- D. reciprocate